

# SPIRE

## *A Contested-Logistics Operating System*

---

SPIRE collapses the data layer underneath military logistics into one role-shaped view. It runs offline on a 2U box in a CONEX, with on-board AI, tamper-evident audit, and one-click coalition release. Built by serving Marines for joint and partnered force employment in degraded, disconnected, intermittent, and limited (DDIL) communications.

**TRL 4. 1st place at Modern Day Marine 2026.**

---

### Four operator surfaces

#### Decision Bridge

The home view. Five live signals on one non-scrolling screen — mission clock and force-protection condition, top alerts, forecasted shortages across Class I / V / VIII / IX, mission-capable rate by unit with seven-day trends, and the health of the audit log itself. A forward operator signs in and reads the unit's posture in three seconds.

#### PULSE

Readiness and risk. Forecasts where shortages will hit and when. Drafts ranked options for the planner — cannibalize, expedite, or cross-level — sorted by which gets readiness back fastest. Predicts equipment failures days ahead and drafts the requisition before the asset breaks. Every action is signed and chained to the audit log.

#### BASTION

Common operating picture. Real geographic map tiles with standard military symbols. Fuses access-control, drone, base-utility, and weather sensor feeds into one threat picture. Multi-sensor incidents correlate automatically. The operator drives the map by typing what they want to see — natural language drives fly-to, filter, query, and overlay.

#### SENTRY

Automated classification. Every record that flows through the system gets a recommended marking with operator override at every step. Sanitized exports for downstream partners. One-click release scoped per partner — FVEY, Japan, Australia, the Philippines — with a unique signature the partner verifies on receipt. The bundle hash is stamped to the audit chain before it leaves the box.

### The platform underneath

#### Universal Ingest

Reads any common file type — CSV, spreadsheets, JSON, XML, fixed-width legacy exports, EDI X12 — from any common source: file drops, secure transfer, email attachments, polled web services, polled databases, streaming feeds. The four core Marine logistics systems are pre-mapped on day one (GCSS-MC asset roster, utilization, service requests; DRRS-MC unit readiness). Extending to another service's system — GCSS-Army, DLA EBS, TC-AIMS — is a one-day engineering task per adapter. When an unfamiliar file arrives, an on-board AI model proposes how to read it; the operator confirms once; the mapping saves so the work isn't redone.

#### Tamper-evident audit

Every change is cryptographically chained at the moment of write. An inspector verifies the chain offline against a published key — no access to the running system required. Feeds existing security monitoring tools (Splunk, ArcSight, QRadar) in standard format.

#### Local-first, no cloud

Fits on a 2U box in a CONEX. Decisions made forward during a SATCOM blackout reconcile automatically when comms restore. Vector-clock sync resolves conflicts without losing forward writes. The on-board AI model lives on the same box as the database.

#### Identity hardening

CAC/PIV authentication scaffolding in place — a single environment-variable flip moves the system from mock auth to production certificate-based auth. Hierarchical role-based access maps to the joint logistics command chain. Records retention compliant with DoD 5015.02. HashiCorp Vault adapter for secrets.

#### Coalition-ready

Five partner release profiles ship pre-built. The redaction engine applies live to every record at release time. The receiving partner verifies the bundle they got matches the signature stamped to the SPIRE audit chain.

### What this lets a force do

A forward operator sees the unit's posture across Class I, V, VIII, and IX on one screen — instead of three browser tabs and a phone call. A planner gets ranked options before the asset breaks, not after the work order opens. A G-4, J-4, or S-4 sees the same canonical shape regardless of which service's data flows through the pipe. A coalition partner gets exactly what their release authority allows,

signed and verifiable. An inspector walks the audit chain offline against a public key. The whole thing runs without satellite, without cloud, on a single 2U box.

## What's in the box

- Four operator surfaces (Decision Bridge, PULSE, BASTION, SENTRY)
- Eight ingest file types × six source channels
- Four pre-mapped Marine logistics systems
- On-board AI for unfamiliar file shapes
- Cryptographically chained audit trail with offline verification
- Common Event Format / syslog export to existing security monitoring
- Five coalition release profiles with live redaction
- CAC/PIV authentication scaffold
- Hierarchical role-based access control
- DoD 5015.02 records retention
- HashiCorp Vault secrets integration
- Air-gap deployment with vector-clock sync on SATCOM restore
- Multi-sensor threat fusion (access-control, drone, base-utility, weather)
- Natural-language operator copilot
- Predictive equipment-failure auto-drafted requisitions
- Monte Carlo demand forecasting (200-path)
- Five-minute install from a clean laptop

Built to IL5 standards (tamper-evident audit, role-based access, smartcard authentication, no cloud egress). Not formally accredited.